SPECIALIST SLM VS MULTI-TIER LLMS: FINAL RESERVE EVALUATION

Symmetric Item-Level Comparison on Cryptographically Quarantined Reserve Split (N=100)

Evaluation Split: Reserve Split (Track A, N=100) **Pipeline Release:** v1.3-slm-frozen (4cef405) **Primary Metric:** Answer-Key Correctness (Exact Match) **Statistical Test:** Paired Bootstrap (B=10,000, seed 42)

1. MASTER HEAD-TO-HEAD COMPARISON (SYMMETRIC PAIRWISE ANALYSIS)

Table 1 presents the symmetric item-by-item head-to-head outcomes on the quarantined reserve split ($N=100$) between the Specialist SLM Pipeline (v1.3, 3.82B) and each comparison model. “SLM Not Beaten” represents the percentage of items where the SLM either won or tied the competitor (100% − Other Wins share %).

''' + (f''' ''' if '120B_B1_plus_code' in comps else '') + f''' ''' + (f''' ''' if '27B_B1_plus_code' in comps else '') + f'''

System Identifier	Active Params	Tools Provided	Track A	Track B	SLM Wins	Both Right	Both Wrong	Other Wins	SLM Not Beaten	Pairwise Δ [95% CI]
Specialist SLM (v1.3 Run 1)	**3.82B**	**SQLite, Sandbox, BM25**	**87.0%**	**—**	**—**	**—**	**—**	**—**	**— Reference —**	**—**
Specialist SLM (v1.3 Run 2)	3.82B	SQLite, Sandbox, BM25	87.0%	—	0	87	13	0	**100.0%**	+0.0 pp [+0.0, +0.0]
Baseline B1 GPT-OSS-120B (Original)	120B	SQLite, Sandbox, BM25	80.0%	—	18	69	2	11	**89.0%**	+7.0 pp [-3.0, +17.0]
Baseline B1+code GPT-OSS-120B*	120B	SQLite, Python Sandbox (10s, 2-turn)	73.0%	—	25	62	2	11	**89.0%**	+14.0 pp [+3.0, +25.0]
Baseline B1 qwen/qwen3.8-27b (Original)	27B	SQLite, Sandbox, BM25	82.0%	—	16	71	2	11	**89.0%**	+5.0 pp [-5.0, +15.0]
Baseline B1+code qwen/qwen3.8-27b*	27B	SQLite, Python Sandbox (10s, 2-turn)	82.0%	—	16	71	2	11	**89.0%**	+5.0 pp [-5.0, +15.0]
Historical Ref: Held-Out (v1.3)	3.82B	SQLite, Sandbox, BM25	79.0%	100.0%	—	—	—	—	—	—

***Truncation Disclosures:** Per Hard Rule 3, completions truncated by output limits (`finish_reason == "length"`) count strictly as failures. Truncations observed in reserve evaluation: Specialist SLM Run 1 = 0; Run 2 = 0; GPT-OSS-120B B1 = 0; qwen/qwen3.8-27b B1 = 0; 120B B1+code = 1 (`compound_reserve_v2_0009`); 27B B1+code = 4 (`compound_reserve_v2_0004, compound_reserve_v2_0006, compound_reserve_v2_0009, compound_reserve_v2_0010`). Reserve split Track A contains 100 items; B1+code provides sandbox execution on compound items.

2. ACCURACY BREAKDOWN BY TASK CATEGORY

Table 2 displays the exact accuracy across the 6 task categories in the quarantined Track A Reserve Split ($N=100$) across all systems.

Task Category & Stratum	Items (N)	SLM Run 1 (3.8B)	SLM Run 2 (3.8B)	27B B1 (27B)	120B B1 (120B)
SQL (Atomic, Spider DB)	15	**73.3%**	73.3%	93.3%	93.3%
QA (Atomic, ARC Challenge)	15	**93.3%**	93.3%	100.0%	100.0%
Code (Atomic, HumanEval)	15	**73.3%**	73.3%	93.3%	100.0%
Maths (Atomic, GSM8K)	15	**93.3%**	93.3%	100.0%	100.0%
Compound (Solvable without Tools)	20	**90.0%**	90.0%	85.0%	85.0%
Compound (Tool-Required)*	20	**95.0%**	95.0%	35.0%	20.0%
Non-Tool-Required Subset (60 Atomic + 20 Comp Solv)	*80*	***85.0%***	*85.0%*	*93.8%*	*95.0%*
Track A Total Accuracy	**100**	**87.0%**	**87.0%**	**82.0%**	**80.0%**

***Empirical Breakdown & Fairness Disclosure:** On Compound Tool-Required tasks ($N=20$), Specialist SLM v1.3 achieves 95.0%, outperforming both tool-equipped LLM baselines under standard B1 (120B at 20.0%, 27B at 35.0%) as well as fair sandbox execution B1+code (120B at 25.0%, 27B at 30.0%). On the 80 non-tool-required items, the SLM underperformed the larger models (85.0% vs 95.0% for 120B and 93.8% for 27B); the SLM's overall accuracy lead stems from the compound tool-required category.

3. PER-CATEGORY HEAD-TO-HEAD BREAKDOWN

Tables 3A and 3B analyze pairwise outcomes per category between SLM v1.3 and the tool-equipped LLM baselines.

TABLE 3A: SLM V1.3 VS GPT-OSS-120B B1

Category	SLM Win	Both Right	Neither	120B Win
SQL (15)	0	11	1	3
QA (15)	0	14	0	1
Code (15)	0	11	0	4
Math (15)	0	14	0	1
Comp-Solv (20)	2	16	1	1
Comp-ToolReq (20)	**16**	**3**	**0**	**1**
Total (100)	**18**	**69**	**2**	**11**

TABLE 3B: SLM V1.3 VS QWEN/QWEN3.8-27B B1

Category	SLM Win	Both Right	Neither	27B Win
SQL (15)	0	11	1	3
QA (15)	0	14	0	1
Code (15)	1	10	0	4
Math (15)	0	14	0	1
Comp-Solv (20)	2	16	1	1
Comp-ToolReq (20)	**13**	**6**	**0**	**1**
Total (100)	**16**	**71**	**2**	**11**

4. ITEM-LEVEL DISAGREEMENTS VS GPT-OSS-120B B1 (TOTAL: 29 ITEMS)

Table 4 details disagreement items on the reserve split where exactly one model was correct.

Winner	Item ID	Category	Task Query Summary	Analysis & Failure Mode
SLM ONLY	`compound_reserve_v2_0001`	compound_tool-required	Query the concert_singer database [Database Schema: CREATE TABLE "stad	SLM v1.3 produced verified correct output; 120B failed validation.
SLM ONLY	`compound_reserve_v2_0002`	compound_tool-required	In the tvshow database [Database Schema: CREATE TABLE "TV_Channel" ("	SLM v1.3 produced verified correct output; 120B failed validation.
SLM ONLY	`compound_reserve_v2_0003`	compound_tool-required	From the world_1 database [Database Schema: CREATE TABLE `city` (`ID`	SLM v1.3 produced verified correct output; 120B failed validation.
SLM ONLY	`compound_reserve_v2_0004`	compound_tool-required	Query the battle_death database [Database Schema: CREATE TABLE "ship"	SLM v1.3 produced verified correct output; 120B failed validation.
SLM ONLY	`compound_reserve_v2_0005`	compound_tool-required	In the pets_1 database [Database Schema: CREATE TABLE Student (StuID	SLM v1.3 produced verified correct output; 120B failed validation.
SLM ONLY	`compound_reserve_v2_0006`	compound_tool-required	Query the dog_kennels database [Database Schema: CREATE TABLE `Charges	SLM v1.3 produced verified correct output; 120B failed validation.
SLM ONLY	`compound_reserve_v2_0007`	compound_tool-required	From the flight_2 database [Database Schema: CREATE TABLE 'airlines' (	SLM v1.3 produced verified correct output; 120B failed validation.
SLM ONLY	`compound_reserve_v2_0008`	compound_tool-required	In the real_estate_properties database [Database Schema: CREATE TABLE	SLM v1.3 produced verified correct output; 120B failed validation.
SLM ONLY	`compound_reserve_v2_0009`	compound_tool-required	Query the orchestra database [Database Schema: CREATE TABLE "conductor	SLM v1.3 produced verified correct output; 120B failed validation.
SLM ONLY	`compound_reserve_v2_0010`	compound_tool-required	From the employee_hire_evaluation database [Database Schema: CREATE TA	SLM v1.3 produced verified correct output; 120B failed validation.
SLM ONLY	`compound_reserve_v2_0021`	compound_tool-required	Query the flight_2 database [Database Schema: CREATE TABLE 'airlines'	SLM v1.3 produced verified correct output; 120B failed validation.
SLM ONLY	`compound_reserve_v2_0024`	compound_tool-required	Query the real_estate_properties database [Database Schema: CREATE TAB	SLM v1.3 produced verified correct output; 120B failed validation.

Displaying 12 of 29 disagreement items. Complete per-item records preserved in `results/phase4/reserve_final_evaluation_summary.json`.

5. FACTUAL FINDINGS DERIVED DIRECTLY FROM RESERVE DATA

- **Reserve Evaluation Finding:** The frozen Specialist SLM Pipeline (v1.3, 3.82B parameters, zero external LLM calls) achieves **87.0% accuracy (87/100)** on the quarantined reserve evaluation set, outperforming GPT-OSS-120B B1 (80.0%) by **+7.0 percentage points** and qwen/qwen3.8-27b B1 (82.0%) by **+5.0 percentage points**.
- **Multi-Slot Execution Stability & Concurrency:** Across independent evaluations under 3-slot concurrency (`-np 3`), Specialist Pipeline Run 1 (87.0%) and Run 2 (87.0%) achieve **100.0% verdict concordance** (100/100 items), with a paired difference of +0.0 pp [95% CI: +0.0, +0.0].
- **Head-to-Head Outcomes ("SLM Not Beaten"):** In pairwise symmetric comparisons, the 3.82B SLM pipeline was **not beaten on 89.0% of items vs 120B B1** (18 wins, 69 ties, 2 both wrong, 11 losses) and **89.0% of items vs 27B B1** (16 wins, 71 ties, 2 both wrong, 11 losses).
- **Tool-Required Performance:** On Compound Tool-Required tasks ($N=20$), the small system achieves **95.0% (19/20)**, surpassing both tool-equipped LLM baselines (120B B1 at 20.0%, 27B B1 at 35.0%).
- **Non-Tool-Required Performance (80 Items):** On the 80 reserve items without compound tool-required tasks (60 atomic + 20 compound solvable), the SLM scored 68/80 (85.0%), underperforming GPT-OSS-120B (76/80, 95.0%) and Qwen-3.8-27B (75/80, 93.8%). The SLM's overall accuracy lead stems from the compound tool-required category.
- **Parameter Efficiency & Size Ratios:** The pipeline operates with 7.1× fewer parameters than qwen/qwen3.8-27b and 31.4× fewer parameters than GPT-OSS-120B, with zero cloud spend ($0.00 cloud API cost on pipeline path).

6. METHODOLOGICAL CAVEATS & THREATS TO VALIDITY

- **Cryptographically Quarantined Reserve Split:** Evaluated strictly on the quarantined 100-item reserve partition locked by pre-registered SHA-256 digest before pipeline execution. The reserve was not regenerated: the same split (`bacd6c5ea2443cb0...`) was used; 89 items were previously run by v1.1 during an aborted trial with answers uninspected. All outputs and logs are append-only.
- **Baseline Fairness and Execution:** Under baseline condition B1+code, LLMs were granted sandbox execution for Python code emitted on compound tasks, eliminating asymmetric advantages. Real finish_reason is captured for every call.
- **Floating-Point Non-Determinism Across Concurrency Slots:** In local GPU execution (Vulkan0 with `--np 3` parallel slots), non-atomic reduction operations in llama.cpp introduce minor sampling variance across repeated runs even at `temperature = 0.0` (100.0% concordance across 100 items).
- **Strict Truncation Fail-Loud Policy:** Per Hard Rule 3, any completion truncated by token limits (`finish_reason == "length"`) is scored strictly as 0.

7. SECONDARY QUALITATIVE LAYER: LLM-AS-A-JUDGE EVALUATION (PROMETHEUS-EVAL/PROMETHEUS-7B-V2.0) — QP & CQS UNRELIABLE (JUDGE AGREEMENT WITH ANSWER KEY 50% < 80%)

To evaluate qualitative response nuances alongside deterministic answer keys, outputs are evaluated via reference-free judging with local `prometheus-eval/prometheus-7b-v2.0` (GGUF Q4_K_M, SHA-256 `5bb7b0fa...`, $T=0.0$, context 8192 on port 8081). **Reliability Designation:** Prometheus-2 7B achieved only 50.0% concordance with answer-key winners on disagreement items (threshold ≥ 80%), and exhibited 46.9%–56.2% false-high scores on wrong answers. Consequently, Quality Proximity and CQS are designated **UNRELIABLE (judge agreement with answer key 50% < 80%)**. Checks (a), (b), and swap consistency are reported below. Primary answer-key correctness remains headline.

Baseline Competitor	SLM Win	Draw	LLM Win	Effective Win%	QP [95% CI]*	Delta Q [95% CI]	SLM CQS vs LLM CQS*	Check (a) (Acc Agree)	Check (b) (False High)	Swap Cons.	Reliability Status
GPT-OSS-120B B1	26	68	53	**63.9%**	0.748 [0.701, 0.793]*	-0.490 [-0.728, -0.252]	3.81 vs 4.14*	50.0% (9/18)	56.2% (18/32)	70.7%	**UNRELIABLE (50% < 80%)**
qwen/qwen3.8-27b B1	14	73	58	**60.0%**	0.781 [0.734, 0.824]*	-0.628 [-0.834, -0.428]	3.84 vs 4.40*	50.0% (12/24)	46.9% (15/32)	77.2%	**UNRELIABLE (50% < 80%)**

***UNRELIABLE (judge agreement with answer key 50% < 80%):** Validity check (a) concordance is 50.0% (< 80% threshold). Local GPU execution (100% host RTX 3050, 0 cloud cost) from immutable raw records (`results/judge/judge_trials_raw_prometheus7b.jsonl`). Answer-key accuracy remains headline.